

Topology of Higher-Order Age-Dependent Random Connection Models

Christian Hirsch, Peter Juhasz

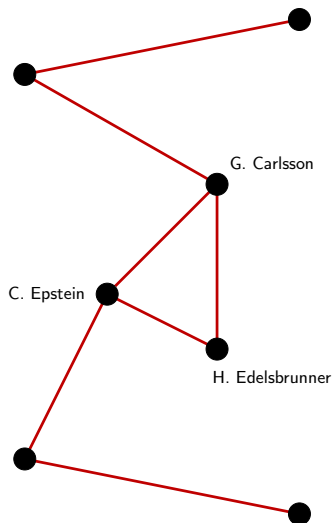


Department of Mathematics
Aarhus University

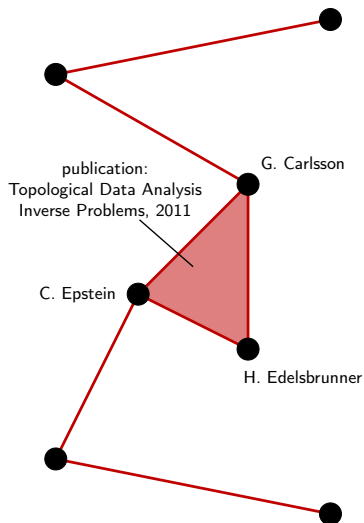
April 13, 2023

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- 3 Age-Dependent Random Connection Model
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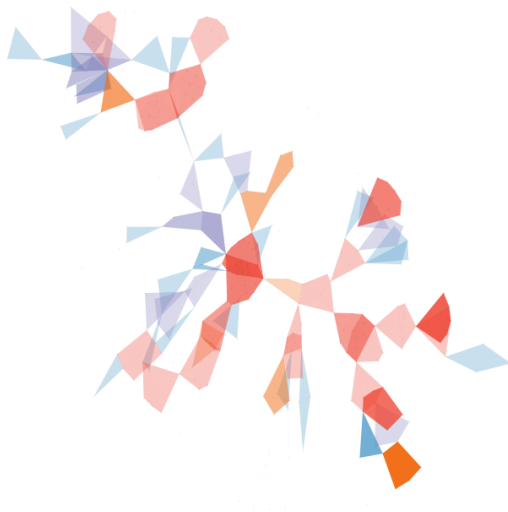
Higher-Order Networks



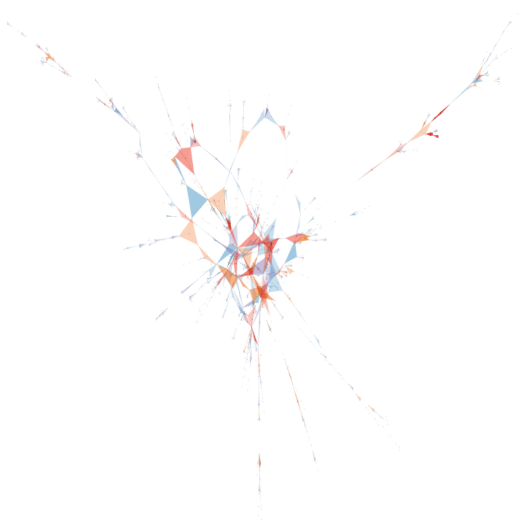
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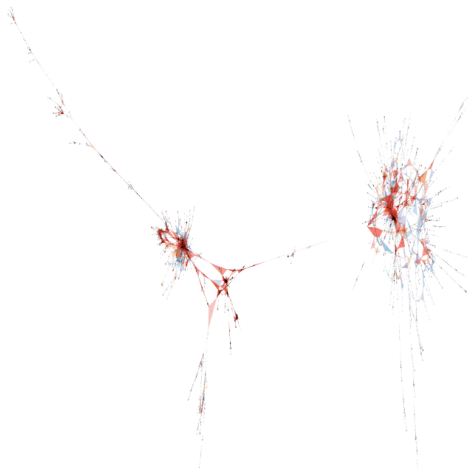
Collaboration of Nature Authors 1990 – 2020



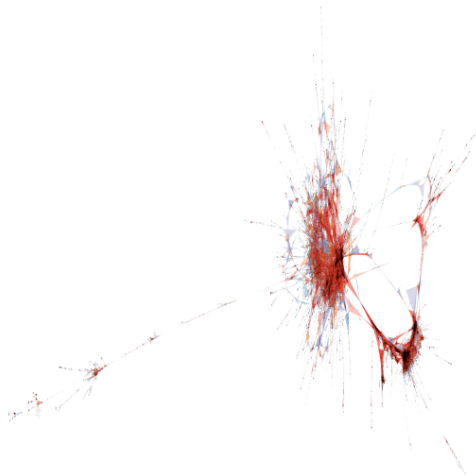
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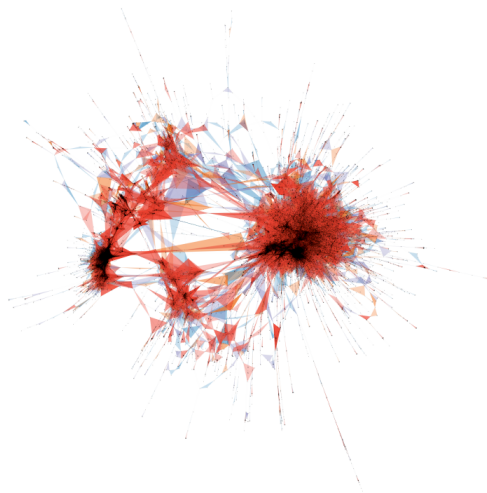
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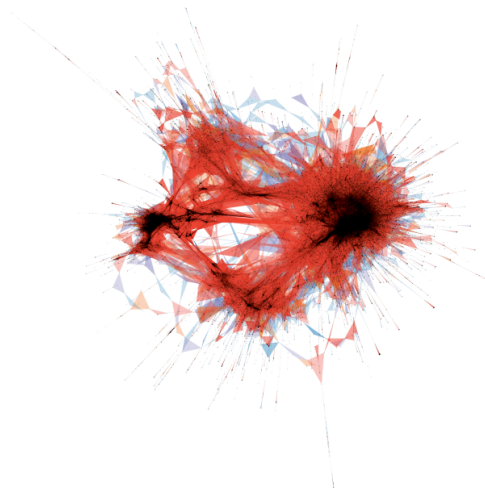
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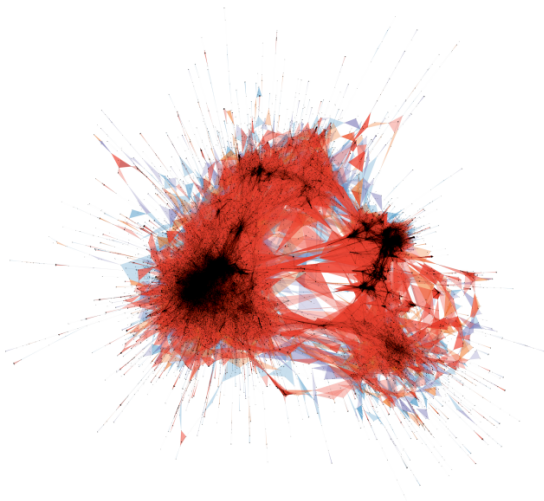
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Goal

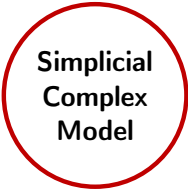


Higher-
Order
Network

Goal

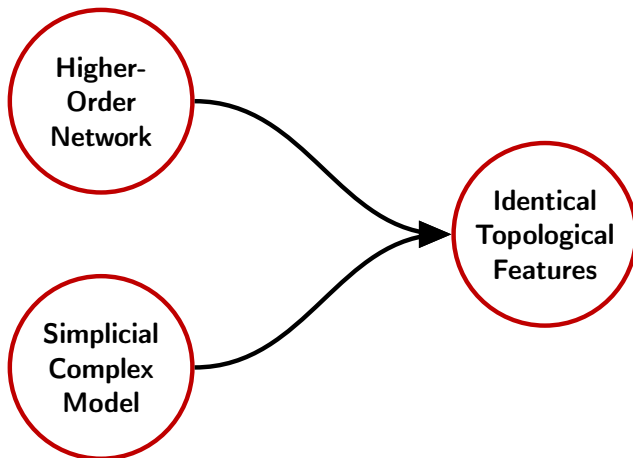


**Higher-
Order
Network**



**Simplicial
Complex
Model**

Goal



Age-Dependent Random Connection Model

Gracar, Peter, et al. "The age-dependent random connection model."
Queueing Systems 93 (2019): 309-331.

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- Vertices: Poisson process

$$\mathcal{P}\{(x_i, t_i)\} \subset \mathbb{R} \times [0, 1]$$

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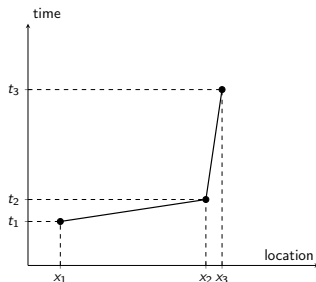
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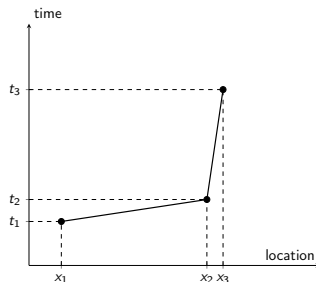
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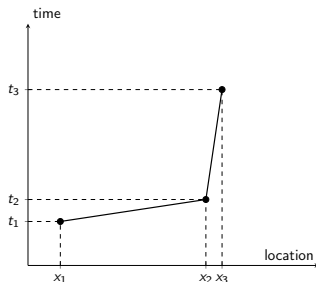
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Properties if $\gamma \in (\frac{1}{2}, 1)$:

- ✓ sparse
- ✓ scale-free
- ✓ ultra small
- ✓ high clustering
- ✗ graphical model



Research Outline

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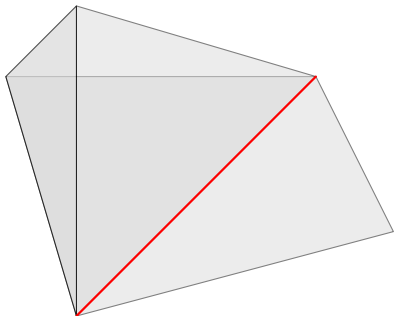
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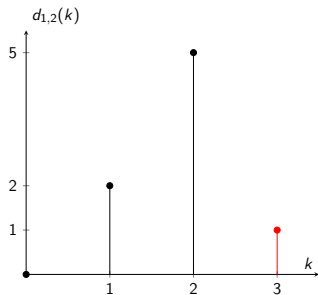
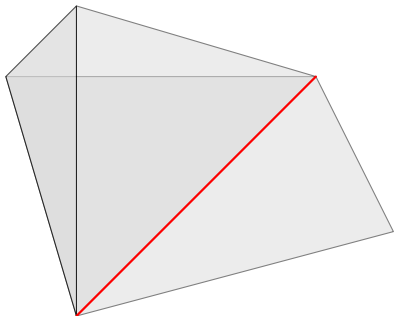
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- Extend ADRCM to match a larger number of characteristics
- Illustrate the findings on simulated networks and on real data

Higher-Order Degree Distributions

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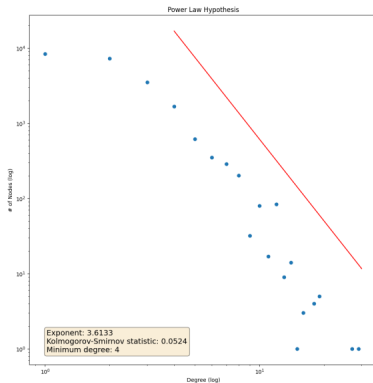
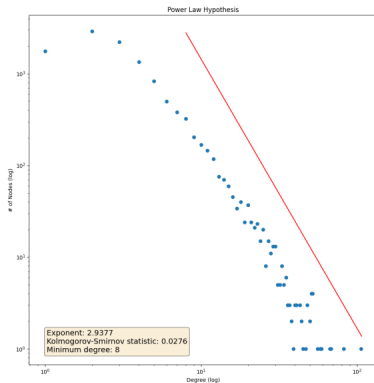


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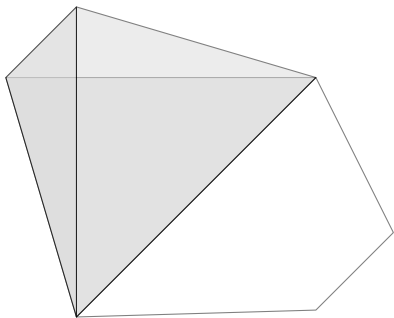


$$\lim_{k \rightarrow \infty} d_{m,m'}(k) \sim k^{m-1-\frac{m+1}{\gamma}}$$

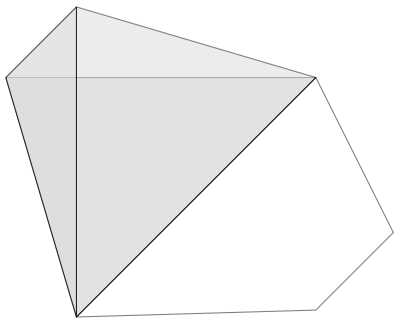
Higher-Order Degree Distribution – Statistics Theory



CLT for Betti Numbers

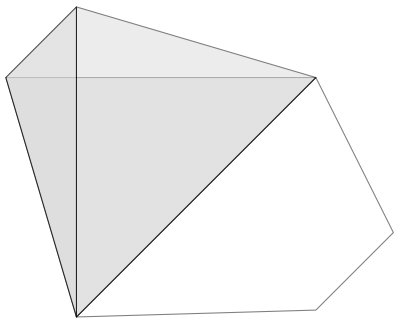


CLT for Betti Numbers



• $\gamma \ll 1$

CLT for Betti Numbers



- $\gamma \ll 1$
- $\frac{\beta_n - \mathbb{E}[\beta_n]}{\sqrt{\text{Var}(\beta_n)}} \xrightarrow{d} \mathcal{N}(0, 1)$

Distribution of Simplex Count

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CLT for Simplex Count

- $\gamma < 1/2$
- S_n : number of m -simplices in the interval $[0, n]$
- $\frac{S_n - \mathbb{E}[S_n]}{\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, \sigma^2)$

Distribution of Simplex Count

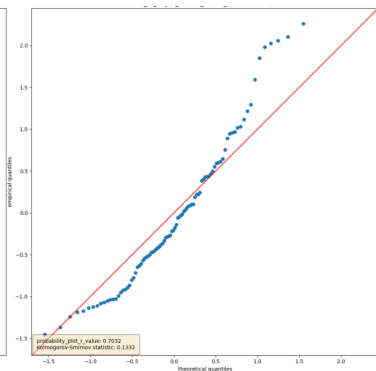
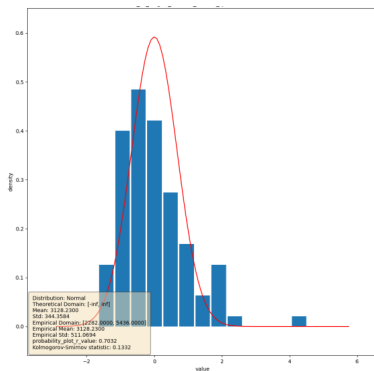
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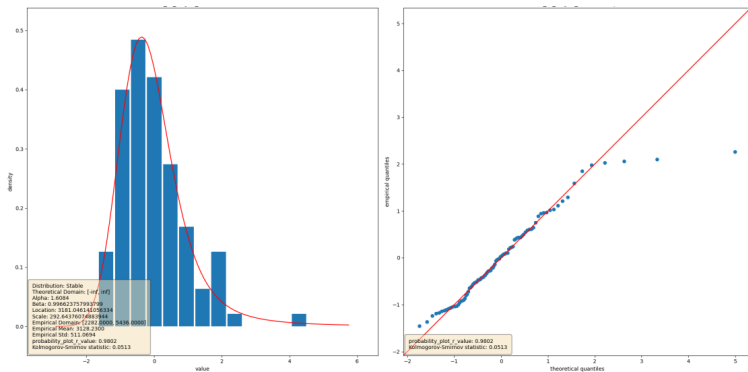
Stable Limit Law for Edge Count

- $1/2 < \gamma < 1$
- S_n : number of edges in the interval $[0, n]$
- $\frac{S_n - \mathbb{E}[S_n]}{n^\gamma} \xrightarrow{d} \mathcal{S}(1/\gamma)$

Stable Distribution of Edge Counts



Stable Distribution of Edge Counts



Thinned Model

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- Goal: match the $d_{0,1}$ and $d_{1,2}$ exponents separately (if $n \rightarrow \infty$)

Thinned Model

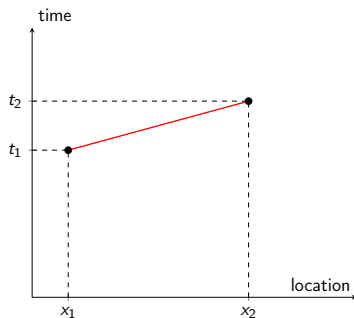
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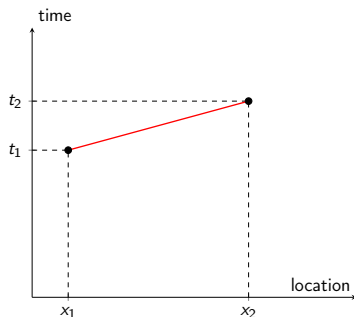
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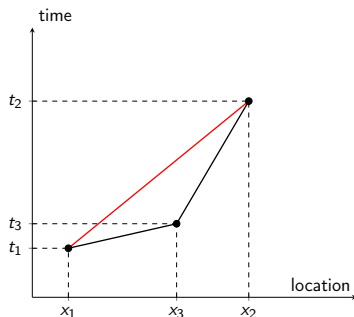
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Protected: $t_1 \approx t_3$

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Q & A